

This Python program is used to **plot different activation functions** used in Neural Networks and Deep Learning.

The program:

1. Imports required libraries.
2. Creates input values from -10 to 10.
3. Defines different activation functions:
 - Linear
 - Sigmoid
 - Tanh
 - ReLU
 - Softmax
4. Plots graphs of all activation functions using Matplotlib.
5. Tests the functions on sample input values.
6. Prints the output of each activation function.

This practical helps understand how activation functions behave mathematically and visually.

Line by Line Explanation

1. Import Libraries

```
import numpy as np
import matplotlib.pyplot as plt
```

Explanation:

- numpy is used for mathematical calculations and array operations.
- matplotlib.pyplot is used to draw graphs and plots.

Viva Answer:

NumPy is used for numerical computations and Matplotlib is used for plotting graphs.

2. Generate Input Values

```
x = np.linspace(-10, 10, 100)
```

Explanation:

- linspace() generates equally spaced values.
- Here:
 - Start = -10
 - End = 10
 - Total values = 100

These values are used as input for activation functions.

Viva Answer:

This line creates 100 input values between -10 and 10 for plotting graphs.

Activation Functions

3. Linear Activation Function

```
def linear(x):  
    return x
```

Explanation:

- Defines a function named linear.
- It returns the same input value.
- No modification is done.

Formula:

$$f(x) = x$$

Example:

If input = 5

Output = 5

Viva Answer:

Linear activation returns the input directly without any change.

4. Sigmoid Activation Function

```
def sigmoid(x):  
    return 1 / (1 + np.exp(-x))
```

Explanation:

- `np.exp(-x)` calculates e^{-x}
- Formula converts input into values between 0 and 1.

Formula:

$$f(x) = \frac{1}{1 + e^{-x}}$$

Characteristics:

- Produces S-shaped curve.
- Used in binary classification.

Example:

Input = 0

Output = 0.5

Viva Answer:

Sigmoid converts any real value into a range between 0 and 1.

5. Tanh Activation Function

```
def tanh(x):  
    return np.tanh(x)
```

Explanation:

- `np.tanh()` calculates hyperbolic tangent.
- Output range is between -1 and 1.

Formula:

$$f(x) = \tanh(x)$$

Characteristics:

- Similar to sigmoid.
- Zero-centered output.

Viva Answer:

Tanh function gives outputs between -1 and 1.

6. ReLU Activation Function

```
def relu(x):  
    return np.maximum(0, x)
```

Explanation:

- `np.maximum(0, x)` compares 0 and x.
- If x is negative → returns 0.
- If x is positive → returns x.

Formula:

$$f(x) = \max(0, x)$$

Example:

- Input = -5 → Output = 0
- Input = 4 → Output = 4

Viva Answer:

ReLU removes negative values and keeps positive values unchanged.

7. Softmax Activation Function

```
def softmax(x):  
    exp_x = np.exp(x - np.max(x))  
    return exp_x / np.sum(exp_x)
```

First Line

```
exp_x = np.exp(x - np.max(x))
```

Explanation:

- `np.max(x)` finds maximum value.
 - Subtracting maximum improves numerical stability.
 - `np.exp()` calculates exponential values.
-

Second Line

```
return exp_x / np.sum(exp_x)
```

Explanation:

- `np.sum(exp_x)` calculates total sum.
- Each value is divided by total sum.
- Final output becomes probability values.

Formula:

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

Viva Answer:

Softmax converts outputs into probability values whose sum is equal to 1.

Plotting Section

8. Create Figure

```
plt.figure(figsize=(10, 8))
```

Explanation:

- Creates graph window.
- Size = width 10 and height 8.

Viva Answer:

This line creates a plotting window with specified size.

9. Linear Graph

```
plt.subplot(3, 2, 1)
plt.plot(x, linear(x))
plt.title("Linear")
plt.grid(True)
```

Explanation:

- `subplot(3,2,1)`:
 - 3 rows
 - 2 columns

- first graph position
- `plot()` draws graph.
- `title()` sets graph title.
- `grid(True)` shows grid lines.

Viva Answer:

This section plots the Linear activation function graph.

10. Sigmoid Graph

```
plt.subplot(3, 2, 2)
plt.plot(x, sigmoid(x))
plt.title("Sigmoid")
plt.grid(True)
```

Explanation:

- Plots Sigmoid graph in second position.
-

11. ReLU Graph

```
plt.subplot(3, 2, 3)
plt.plot(x, relu(x))
plt.title("ReLU")
plt.grid(True)
```

Explanation:

- Plots ReLU graph.
-

12. Tanh Graph

```
plt.subplot(3, 2, 4)
plt.plot(x, tanh(x))
plt.title("Tanh")
plt.grid(True)
```

Explanation:

- Plots Tanh graph.
-

13. Softmax Graph

```
plt.subplot(3, 2, 5)
plt.plot(x, softmax(x))
plt.title("Softmax")
plt.grid(True)
```

Explanation:

- Plots Softmax graph.
-

14. Adjust Layout

```
plt.tight_layout()
```

Explanation:

- Adjusts spacing between graphs automatically.

Viva Answer:

It prevents overlapping between graphs.

15. Display Graphs

```
plt.show()
```

Explanation:

- Displays all plotted graphs on screen.

Viva Answer:

plt.show() is used to display graphs.

Input Array Section

16. Create Sample Array

```
arr = np.array([-1, 0, 1])
```

Explanation:

- Creates NumPy array with sample values.
-

17. Print Input Array

```
print("Input Array:", arr)
```

Explanation:

- Displays input values.
-

18. Sigmoid Output

```
print("Sigmoid Output:")  
print(sigmoid(arr))
```

Explanation:

- Applies sigmoid function on array.
 - Prints result.
-

19. Tanh Output

```
print("Tanh Output:")  
print(tanh(arr))
```

Explanation:

- Applies tanh function.
-

20. ReLU Output

```
print("ReLU Output:")  
print(relu(arr))
```

Explanation:

- Applies ReLU function.
-

21. Softmax Output

```
print("Softmax Output:")  
print(softmax(arr))
```

Explanation:

- Applies softmax function.
 - Prints probability outputs.
-

Important Viva Questions with Answers**Q1. What is an activation function?**

An activation function decides whether a neuron should be activated or not.

Q2. Why are activation functions important?

They introduce non-linearity into neural networks.

Q3. Which activation function is most commonly used?

ReLU is most commonly used.

Q4. What is the output range of sigmoid?

0 to 1.

Q5. What is the output range of tanh?

-1 to 1.

Q6. Why is softmax used?

Softmax is used for multiclass classification problems.

Q7. What is ReLU?

ReLU returns 0 for negative values and x for positive values.

Q8. Which library is used for plotting?

Matplotlib library.

Q9. Which library is used for numerical operations?

NumPy library.

Q10. What does plt.show() do?

It displays the plotted graphs.